

Introduction

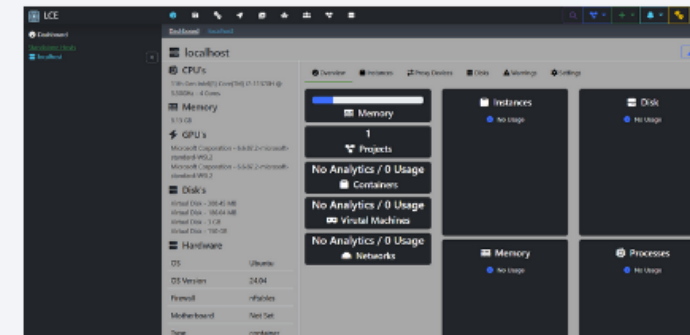
The artifact describes a lightweight compute environment that streamlines the creation of system containers by abstracting complex configuration steps. It leverages the LXD API to provision and manage containers efficiently, ensuring consistent and scalable operations. The overall design aligns with a Platform-as-a-Service model, allowing users to deploy and manage workloads with minimal overhead.

Service-oriented Architecture Using System Container

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Aim

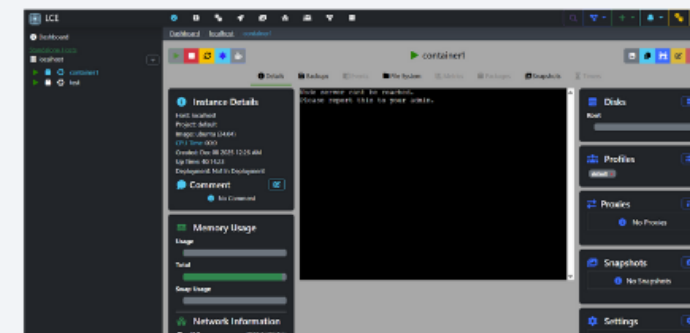
The primary focus of this research is to identify how system containers such as LXC can be used to deploy services of choice following Service-Oriented Architecture model.

Technology Stack

- Php v8.4
- LXD v5.21.4 LTS
- LXC v5.21.4 LTS
- Node v24.11.1
- Composer v2.9.2
- MariaDB v10.11.13

Key Features

- User can choose pre-deployment specifications,
- Users can create containers,
- Users can host applications of their choice,
- Users can start and stop the services,
- Users can revoke the containers,



Problem Statement

The problem centers on the growing complexity involved in deploying and managing servers, which demands significant time and expertise. Despite the maturity of virtualization and containerization technologies, there is still no focused solution dedicated to simplifying operations within the system-container domain. This gap creates operational overhead and limits organizations seeking lightweight, efficient infrastructure management.

Conclusion

The creation of the Lightweight Compute Environment marks a deliberate effort to simplify how system containers are managed in modern infrastructure. Throughout the project, an agile framework guided each iteration, allowing steady refinement of features and workflows. This approach ensured that feedback and evolving requirements were integrated without disrupting progress. The system relies on the LXD API to control and orchestrate system containers, giving it a reliable foundation for scalable operations. By incorporating the LXC container system, the environment maintains strong performance while preserving low resource overhead. Each component was selected to reduce the friction often associated with container lifecycle tasks. The resulting architecture offers a streamlined experience that minimizes manual configuration. This shift supports teams seeking predictable and efficient container management. The artifact ultimately provides a practical pathway to operational simplicity. Its implementation demonstrates how thoughtful design can ease the daily demands of running system containers.

References

- Rajan, A.P., 2013. Evolution of cloud storage as cloud computing infrastructure service. arXiv preprint arXiv:1308.1303.
- Kaur, R. and Kaur, A., 2014. A review paper on evolution of cloud computing, its approaches and comparison with grid computing. International Journal of Computer Science and Information Technologies, 5(5), pp.6060-6063.